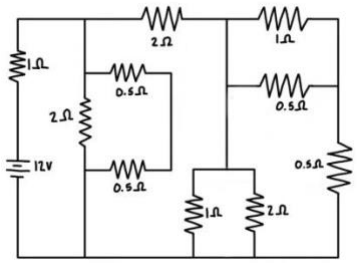
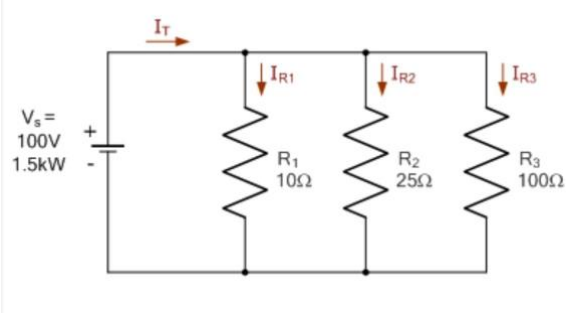
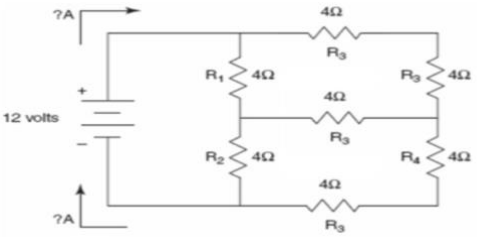
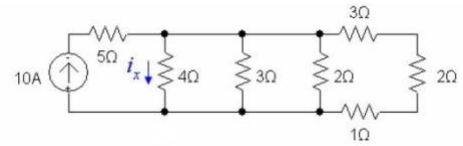
ADITYA UNIVERSITY

Department of freshman Engineering

Important Questions

Course name: Basic Electrical and Electronic Engineering **Course code:** 2501EE01

UNIT-1					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Find the Equivalent resistance & total current to the given circuit. 	L3	CO1	4M
	b	What is Power factor? What is the Significant.	L2	CO1	4M
2	a	Verify the KCL to the given circuit 	L3	CO1	4M
	b	Define KCL & KVL Explain with a simple numerical	L2	CO1	4M
3	a	Find the Equivalent resistance & total current to the given circuit. 	L3	CO1	4M
	b	Compare AC & DC Supply	L3	CO1	4M

4	a	Find i_x to the given circuit. 	L3	CO1	4M
	b	Define (i) Time Period (ii) frequency (iii)Amplitude (iv)peak to peak value	L1	CO1	4M
5	a	$I(t) = 2 \sin(5t - 45^0)$ Find Magnitude, Frequency, Time Period, RMS Value, Average value & Phase Angle	L3	CO1	4M
	b	Explain Power Triangle & Define all the terms in the power triangle.	L2	CO1	4M
6	a	$V(t) = 10 \cos(4t + 45^0)$ Find Magnitude, Frequency, Time Period, RMS Value, Average value & Phase Angle	L3	CO1	4M
	b	Define & Derive RMS Value and Average Value	L3	CO1	4M

UNIT-2

QNO	Sub QNO	Question	BT	CO	Marks
1	a	What is an Electrical Machine? Briefly explain the conversion of an electrical machine from one form to another form with principle.	L2	CO2	4M
	b	Compare the Operational difference between Transformer & Induction motor.	L3	CO2	4M
2	a	Classify Different type of Measuring Instruments to measure AC & DC Quantities with basic principle.	L2	CO2	4M
	b	Classify different Resistors? How to Measure High Resistance? Explain with a neat sketch.	L3	CO2	4M
3	a	What is the Difference between Back emf & Generated emf? State the function of each & Explain Briefly.	L2	CO2	4M
	b	Why Single Phase Induction Motors are not Self Started? Explain Any two types of Single Phase Induction Motors with neat diagrams.	L3	CO2	4M
4	a	What is the Role of Transformer in Electrical Power System? Explain the operation of Transformer with neat diagram.	L2	CO2	4M
	b	How to Find the direction of Mechanical & Electrical output of the Machines & Explain Principle of each.	L3	CO2	4M
5	a	Describe the Role of Commutator in DC Machines? Explain.	L2	CO2	4M
	b	How to Find the Medium Resistor? Explain the principle & Derive the expression for the unknown Resistor.	L2	CO2	4M
6	a	Draw the neat sketch the PMMC ? Explain the Operation.	L2	CO2	4M
	b	Compare Difference type of MI types of Instruments & Explain their Principles.	L3	CO2	4M